

計畫名稱：高濃度二相懸浮流體的先進數值計算：非均值顆粒與主動式學習跨尺度計算

計畫編號：NSTC 112-2221-E-002-186-MY3

主持人：周逸儒

執行單位：國立臺灣大學應用力學所

參與人員：蔡曜鴻、周逸儒

On the suspension and deposition within turbidity currents

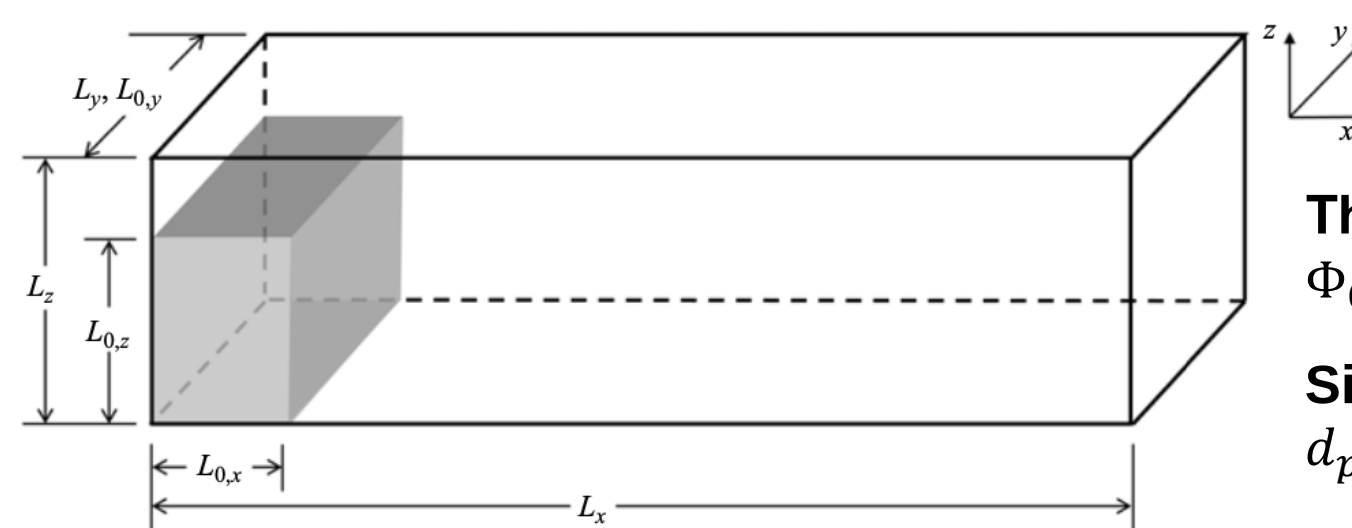
Yao-Hung Tsai¹ and Yi-Ju Chou^{1,2}¹Institute of Applied Mechanics, National Taiwan University, Taipei 10617, Taiwan²Ocean Center, National Taiwan University, Taipei 10617, Taiwan

Abstract

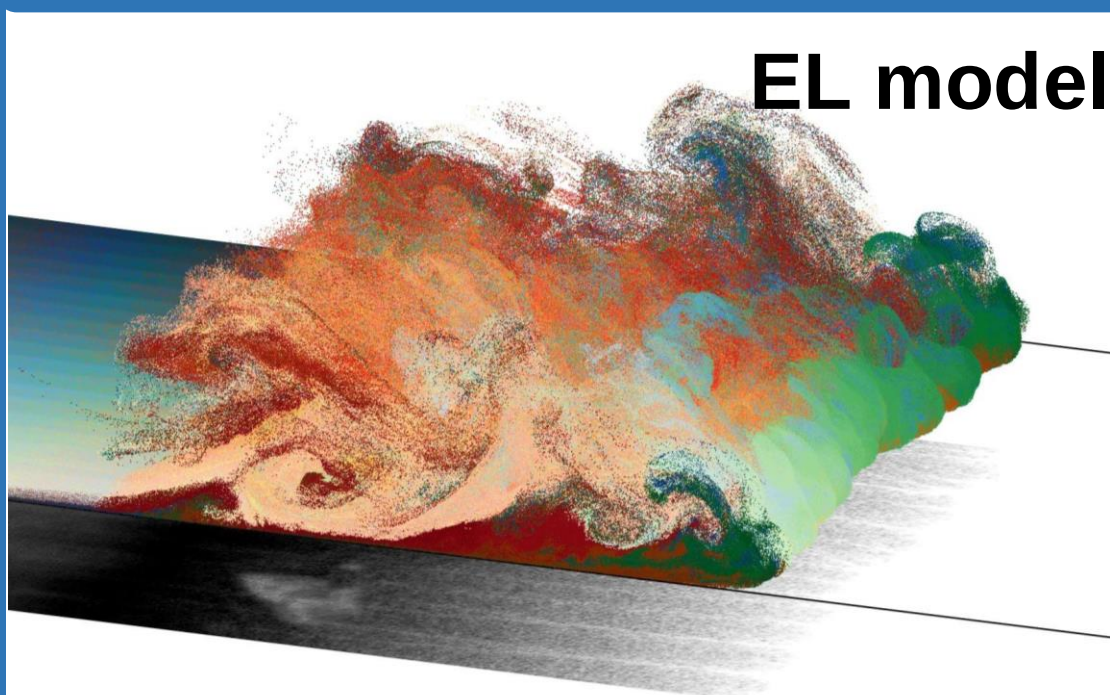
Numerical simulations investigate particle suspension and deposition within turbidity currents. Using Lagrangian particle tracking and a discrete element model, our approach examines autosuspension, deposition, and bulk turbidity current behavior, focusing on flow regimes where particle settling and buoyancy-induced hydrodynamics are equally significant. We first analyze the turbidity current's main body formed by suspended particles, revealing its temporal evolution through slumping, propagation, and dissipation stages. We then explore particle deposition, emphasizing transverse and longitudinal variations. Transverse variations stem from lobe-and-cleft (LC) flow structures, while longitudinal variations result from vortex detachment, particularly pronounced for larger particles. As particle size increases, leading to a Stokes number above 0.1, rapid settling suppresses LC flow structures, creating wider lobes at the deposition height. This study advances the understanding of turbidity current dynamics and particle transport mechanisms and has been published in *Journal of Fluid Mechanics* (Tsai et al. 2025).

Methods

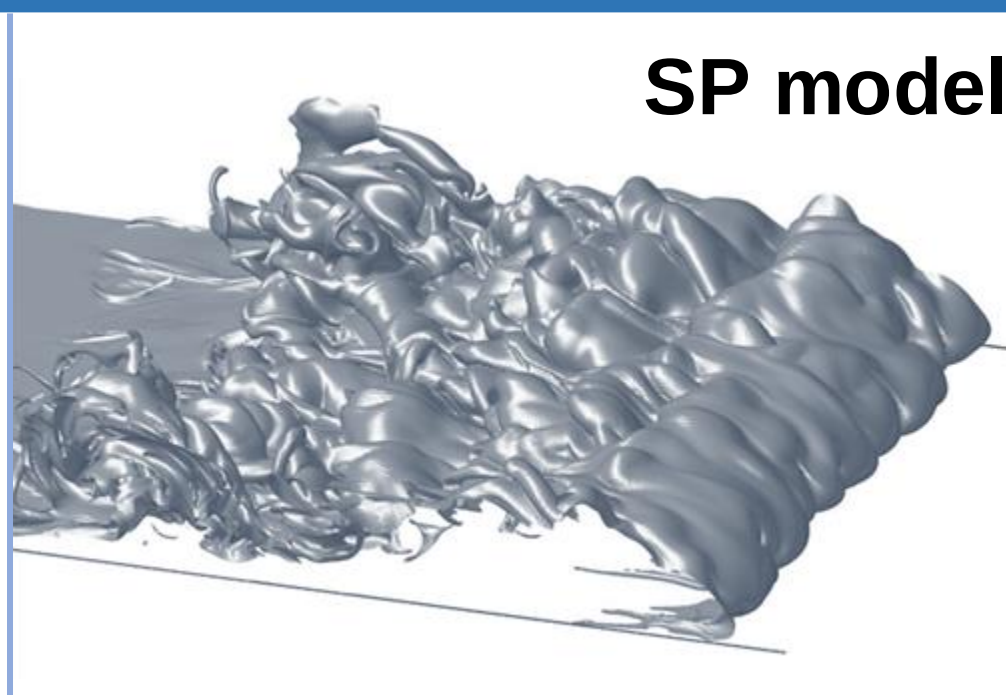
The Euler-Lagrangian model is a numerical method for simulating fluid and particle interactions. The fluid phase is solved on an Eulerian grid using the Navier-Stokes equations, while the particle phase is treated as point particles, governed by Newton's second law, with a particle grid method handling interactions. Direct numerical simulation (DNS) is employed to compute flow dynamics, allowing for particle trajectory tracking and flow field data extraction within each grid cell.

Three particle volume fractions
 Φ_0 : 0.005, 0.01, 0.02Six particle sizes
 d_p : 53, 65, 75, 84, 99, 112 μm

Results

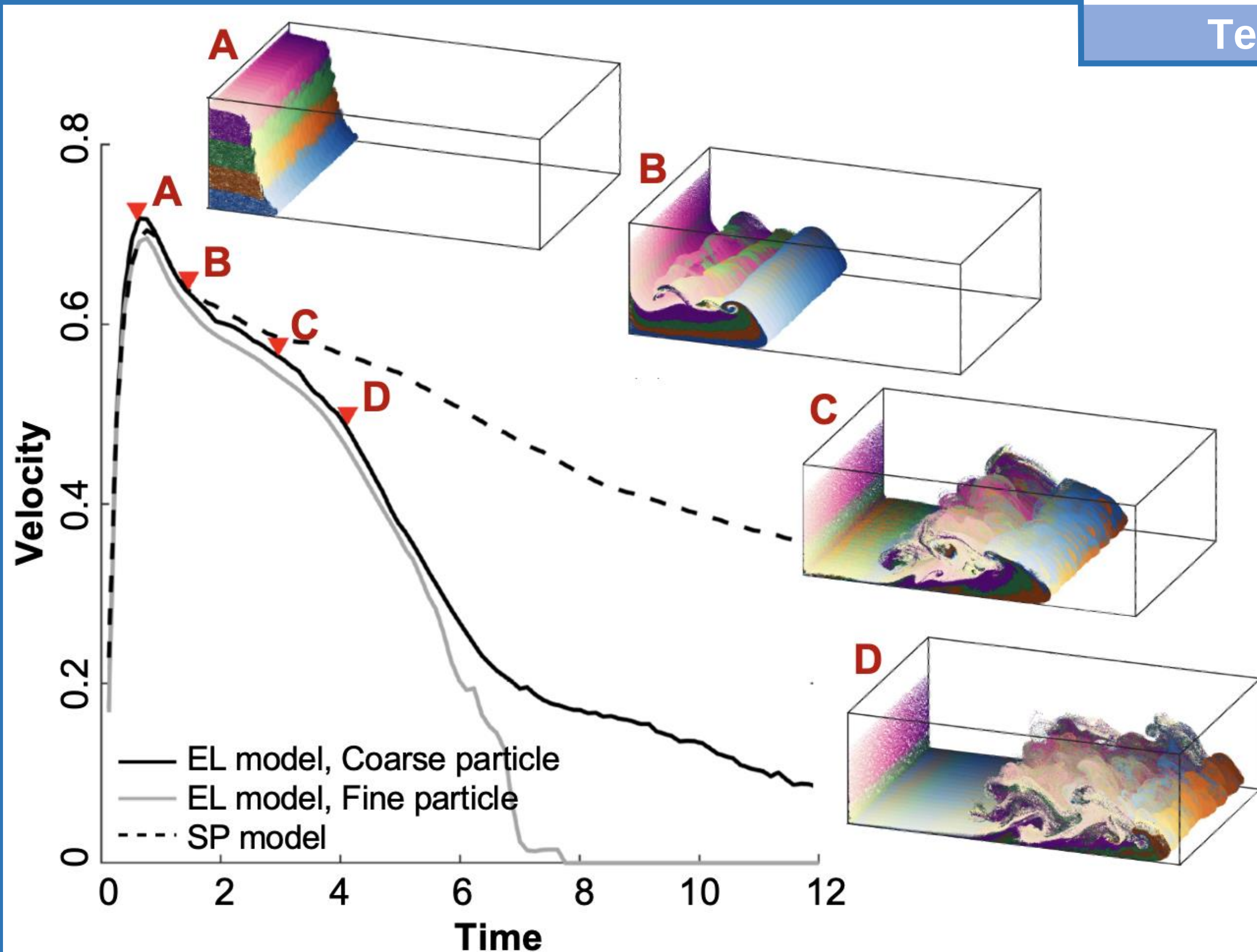


EL model



SP model

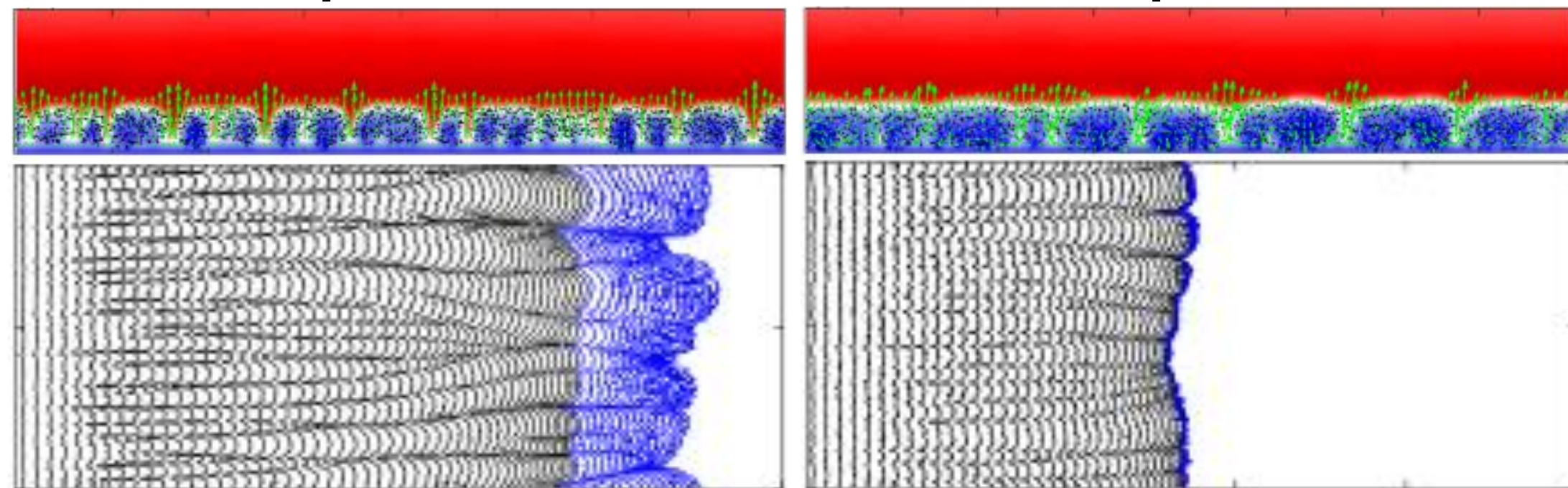
The evolution of turbidity currents



Time histories of the front speed

Fine particle

Coarse particle



Temporal evolution of the deposition contour

Fine particle

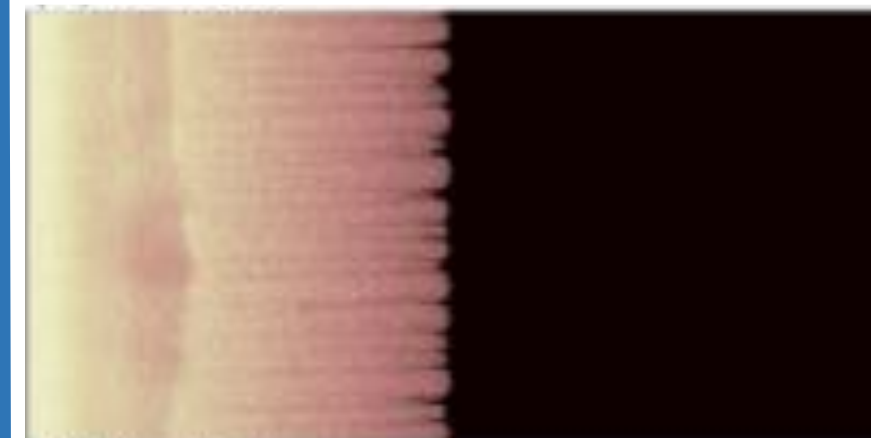
Time = 1.3



Time = 5.1



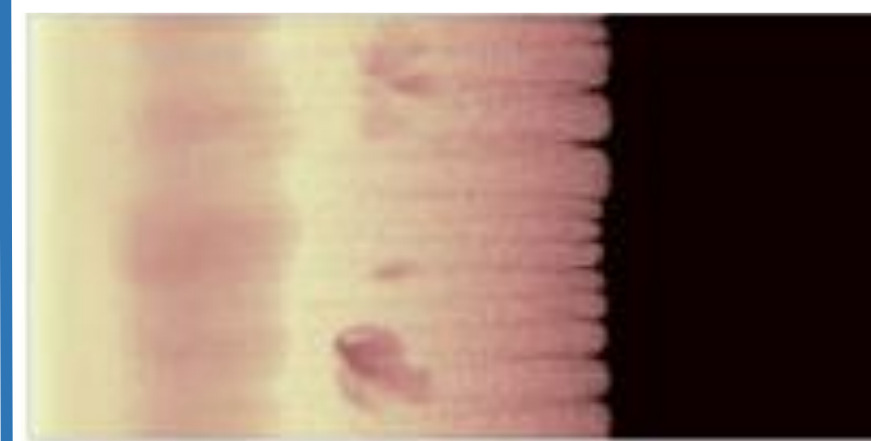
Time = 2.5



Time = 6.4



Time = 3.8



Time = 8.9



Deposition height at different time

Reference

- NECKER, F., HARTEL, C., KLEISER, L. & MEIBURG, E. 2002 High-resolution simulations for particle-driven gravity currents. *Intl J. Multiphase Flow*. 28, 279–300.
- XIE, J., HU, P., ZHU, C., YU, Z. & PÄHTZ, T. 2023 Turbidity currents propagating down an inclined slope: particle auto-suspension. *J. Fluid Mech.* 954, A44.
- YH, T., YJ Chou., 2025 On the suspension and deposition within turbidity currents. *J. Fluid Mech.* 1003, A1.

Conclusion

- This study examines the progression of turbidity currents and the role of the lobe-and-cleft (LC) structure in particle deposition.
- This is the first study to apply Lagrangian particle simulations to turbidity currents at $Re \sim O(1000)$, revealing particle suspension and deposition.